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ST4003CEM Object Oriented Programming

Prabisha Khadka

1. Write a program that prints Hello World! to the console. Save the program as Hello.java.

```
Hello1.java > Hello1
public class Hello1 {
    Run | Debug
    public static void main(String[] args) {
        System.out.println(x: "Hello World!");
    }
}
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Hello World!
PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>
```

2. Extend/Modify the above program to print two additional lines:

   This is my first program

   I am on module ST4003CEM

   System.out.print (not println) and produces the same output.

```
Hello1.java > Hello1
public class Hello1 {
    Run | Debug
    public static void main(String[] args) {
        System.out.print(s: "Hello World!\n");
        System.out.print(s: "This is my first program\n");
        System.out.print(s: "I am on module ST4003CEM\n");
    }
}
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>  & 'C:\Program Files\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp
' 'C:\Users\OMEN\AppData\Roaming\Code\User\workspaceStorage\cc104f76c3e93f02eda6cd0a86ddf96b\redhat.java\jdt_ws\Java_e473238e\bin' 'Hello1'
Hello World!
This is my first program
I am on module ST4003CEM
PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>
```

3. Write a program that prints the following output:

   A "quoted" String is

   'much' better if you learn

   the rules of "escape sequences."

   Also, "" represents an empty

   String. Don't forget: use \"

   instead of " !

   '' is not the same as "

```java
EscapeSequencesDemo.java > EscapeSequencesDemo
public class EscapeSequencesDemo {
    Run | Debug
    public static void main(String[] args) {
        System.out.println(x: "A \"quoted\" String is");
        System.out.println(x: "'much' better if you learn");
        System.out.println(x: "the rules of \"escape sequences.\"");
        System.out.println(x: "Also, \"\" represents an empty");
        System.out.println(x: "String. Don't forget: use \\\" ");
        System.out.println(x: "instead of \" !");
        System.out.println(x: "'' is not the same as \"");
    }
}
```

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>  & 'C:\Program Files\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp
' 'C:\Users\OMEN\AppData\Roaming\Code\User\workspaceStorage\cc104f76c3e93f02eda6cd0a86ddf96b\redhat.java\jdt_ws\Java_e473238e\bin' 'EscapeSeque
ncesDemo'
A "quoted" String is
'much' better if you learn
the rules of "escape sequences."
Also, "" represents an empty
String. Don't forget: use \"
instead of " !
'' is not the same as "
PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>
```

4. WAP that prints the following pattern:

   *

   **

   ***

   ****

   *****

```java
StarPattern.java > StarPattern
public class StarPattern {
    Run | Debug
    public static void main(String[] args) {
        // Loop from 1 to 5
        for (int i = 1; i <= 5; i++) {
            // Print i stars on each line
            for (int j = 1; j <= i; j++) {
                System.out.print(s: "*");
            }
            // Move to the next line
            System.out.println();
        }
    }
}
```

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>  & 'C:\Program Files\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp
' 'C:\Users\OMEN\AppData\Roaming\Code\User\workspaceStorage\cc104f76c3e93f02eda6cd0a86ddf96b\redhat.java\jdt_ws\Java_e473238e\bin' 'StarPattern
'
*
**
***
****
*****
PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>
```

5. WAP prints the following pattern:

```
*********

*********

*********

*********

*********
```

```
StarRectangle.java > StarRectangle
public class StarRectangle {
    Run | Debug
    public static void main(String[] args) {
        // Outer loop controls the number of rows (5 rows)
        for (int i = 1; i <= 5; i++) {
            // Inner loop prints 9 stars in each row
            for (int j = 1; j <= 9; j++) {
                System.out.print(s: "*");
            }
            // Move to the next line after each row
            System.out.println();
        }
    }
}
```

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>  & 'C:\Program Files\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp
' 'C:\Users\OMEN\AppData\Roaming\Code\User\workspaceStorage\cc104f76c3e93f02eda6cd0a86ddf96b\redhat.java\jdt_ws\Java_e473238e\bin' 'StarRectang
le'
*********
*********
*********
*********
*********
PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>
```

6. WAP that prints the following output:

A well-formed Java program has a main method with { and } braces.

A System.out.println statement has (and) and usually a String that starts and ends with

a " character.

(But we type \" instead!)

What is the difference between

a ' and a "? Or between a " and a \"?

One is what we see when we're typing our program. The other is what appears on the

"console."

EscapeSequenceExample.java > EscapeSequenceExample

```java
public class EscapeSequenceExample {
    Run | Debug
    public static void main(String[] args) {
        System.out.println("A well-formed Java program has a main method with { and } braces.");
        System.out.println("A System.out.println statement has ( and ) and usually a String that starts and ends with");
        System.out.println("a \" character.");
        System.out.println("(But we type \\\" instead!)");
        System.out.println("What is the difference between");
        System.out.println("a ' and a \"?  Or between a \" and a \\\"?");
        System.out.println("One is what we see when we're typing our program. The other is what appears on the");
        System.out.println("\"console.\"");
    }
}
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>  & 'C:\Program Files\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp
' 'C:\Users\OMEN\AppData\Roaming\Code\User\workspaceStorage\cc104f76c3e93f02eda6cd0a86ddf96b\redhat.java\jdt_ws\Java_e473238e\bin' 'EscapeSeque
nceExample'
A well-formed Java program has a main method with { and } braces.
A System.out.println statement has ( and ) and usually a String that starts and ends with
a " character.
(But we type \" instead!)
What is the difference between
a ' and a "?  Or between a " and a \"?
One is what we see when we're typing our program. The other is what appears on the
"console."
PS C:\Users\OMEN\OneDrive\Documents\Desktop\Java>
```